

SMART INDIA HACKATHON 2026



SmartNAWI – Digital Testing & Compliance Platform for NAWI



- **Problem Statement ID** – SIH26035
- **Problem Statement Title** – Development of a Software Program / Application for Generation of Test Reports for Non-Automatic Weighing Instruments (NAWI) as per OIML Recommendation R-76
- **Theme** – Miscellaneous
- **PS Category** – Software
- **Team ID** – SIH26_13
- **Team Name (Registered on portal)** – Team Limitless
- **Organization** – Ministry of Consumer Affairs, Food & Public Distribution
- **Department** – Department of Consumer Affairs (DoCA)

Digital data recording, compliance evaluation and standardized test report generation for NAWI model approval.



Team
Limitless

SmartNAWI – Digital Testing & Compliance Platform for NAWI

❖ Proposed Solution

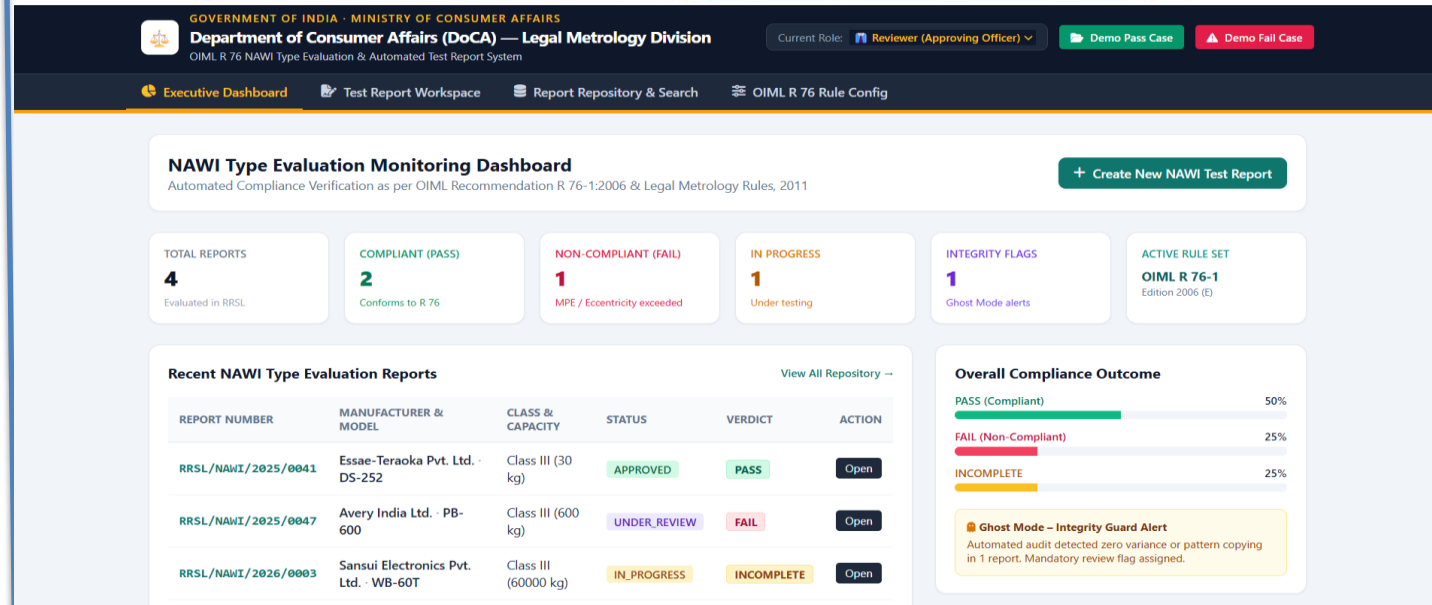
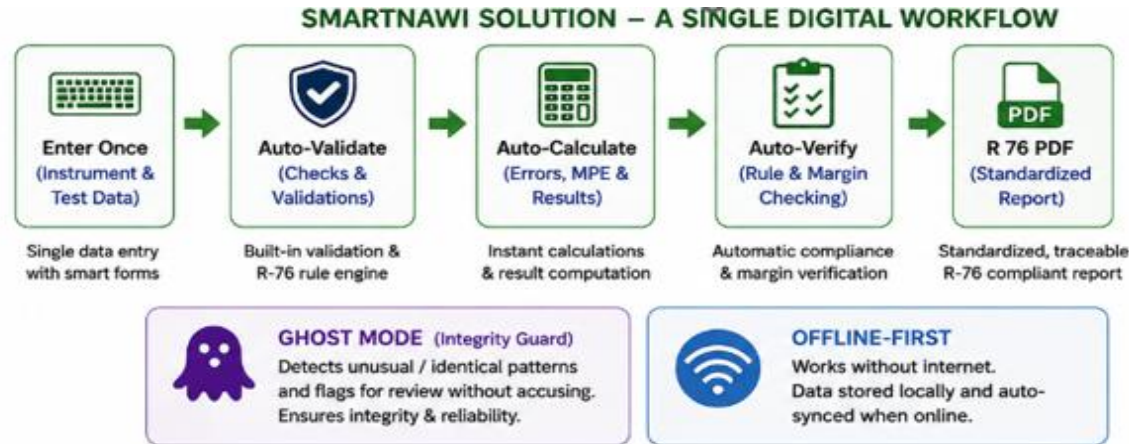
From Manual Testing → SmartNAWI



WHY CURRENT PROCESS IS INEFFICIENT?

- ✗ Same data entered multiple times
- ✗ Calculations verified manually
- ✗ Reports prepared separately
- ✗ Historical records hard to retrieve

➤ OUR SOLUTION — Smart NAWI



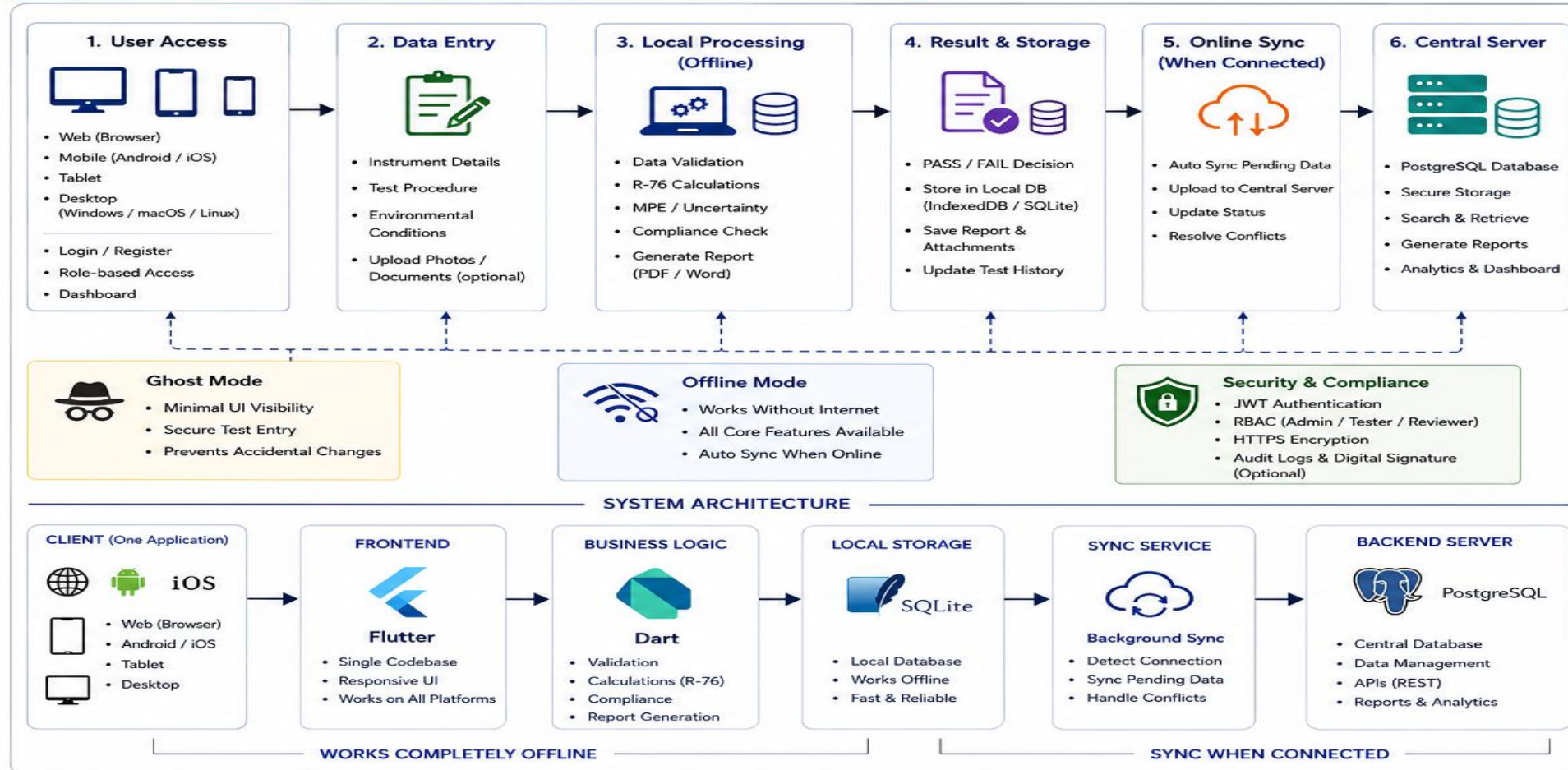
HOW SMARTNAWI WORKS

SmartNAWI transforms traditional NAWI testing into a unified digital workflow. Test data is captured once, automatically validated against OIML R76 requirements, processed through a rule-based compliance engine, and converted into standardized digital reports. The platform ensures consistency, traceability, audit readiness, and faster decision-making throughout the evaluation process.

✓ WORKING PROTOTYPE

- Automated OIML R76 compliance verification
- Rule-based PASS / FAIL evaluation
- Standardized digital report generation
- Dashboard-based monitoring & tracking
- Searchable audit-ready records
- End-to-end digital traceability

One Structured Pipeline – Data entered once, everything after that is automatic



END TO END SECURITY

JWT Authentication

RBAC

HTTPS / SSL

Audit Logs

Encryption

Digital Signature (Optional)

Existing standard. Proven testing. Practical digital implementation for NAWI type evaluation and compliance.

01 FEASIBILITY

Can we build it?

- ✓ **Standard already exists**
OIML R 76 defines NAWI test procedures used during type evaluation, including maximum permissible error (MPE) and reporting requirements.
- ✓ **Technology already exists**
Web stack + relational database; no specialised hardware needed for the software layer.
- ✓ **Deterministic by design**
Same observation + same rule → same calculation → same verdict.
- ✓ **Traceable output**
Observation → calculation → rule → PASS/FAIL → report.

02 VIABILITY

Will it work beyond the demo?

- ✓ **Less manual work**
One-time data entry; automated calculations, validation, compliance verification and standardized report generation.
- ✓ **Consistent decisions**
Rule-based logic reduces calculation and transcription variability.
- ✓ **Easy to maintain**
R-76 rules can live in a configurable layer rather than hard-coded UI logic.
- ✓ **Ready to scale**
Same workflow can support test centres, reviewers and searchable digital records.

03 ENGINEERING VALIDATION

SMARTNAWI ENGINEERING BASIS

Rule-driven architecture for legally relevant calculations

RULE LAYER

R-76 clauses mapped to configurable test rules

VALIDATION LAYER

Input → unit → MPE → PASS/FAIL validation gates

EVIDENCE LAYER

Observation → rule ID → calculation → verdict → report

WHY THIS MATTERS

No black-box model decides the metrology verdict; every result remains explainable.

DESIGN PRINCIPLE

Automate the workflow — keep the metrology logic explicit.

Design basis: OIML R 76 + OIML D 31 + WELMEC software guidance

DETERMINISTIC = same input → same result

EXPLAINABLE = every verdict has a rule path

TRACEABLE = every report has an evidence trail

What changes in the field when NAWI testing and model approval documentation become digital

Faster standardized test reports • fewer transcription errors • stronger traceability

1 REAL-WORLD IMPACT



MANUAL TODAY

Repeated data entry
Hand calculations
Paper / folders
Tester-dependent

SMARTNAWI

Enter once per test
Automatic calculations
Searchable digital history
Rule-based & consistent

2 WHO BENEFITS

TESTERS / LABS

Less re-entry • faster test report preparation •
fewer transcription steps

TEST CENTRES / INDUSTRY

Standardized test reports • searchable history •
easier review

REGULATORS / DoCA

Compliance verification • audit trail •
consistent evidence

**OFFLINE-FIRST • TEST WITHOUT
INTERNET**

**GHOST MODE • DISTRACTION-FREE
ENTRY**

3 RESEARCH EVIDENCE

OIML Bulletin • 2025

Automated / remote OIML R 76
testing compared with traditional
manual testing.

3–150 kg
tested range

$E_n < 1$
all load points

Automated / remote results were
highly consistent with the traditional
manual method.

Digital testing can reduce manual handling
while preserving an evidence trail.

Source: Yang et al., OIML Bulletin LXVI(3), 2025

LESS RE-ENTRY • BETTER TRACEABILITY • OFFLINE OPERATION • STANDARDISED REPORTING • RESEARCH-BACKED

What international software guidance and the current R-76 revision direction support

Standards, software governance and evidence traceability for NAWI testing

1 STANDARDS & LEGAL BASIS

OIML R 76-1

Metrological & technical requirements + standardized testing for NAWIs.

[Source ↗](#)

OIML R 76-2

Evaluation/report structure, test results and maximum permissible error notation.

[Source ↗](#)

INDIA — LEGAL METROLOGY

DCA framework provides GATCs; Class III/IIII NAWIs up to 150 kg are listed for GATC verification. RRSs support traceability of standards.

[Source ↗](#)

2 SOFTWARE & GOVERNANCE

WELMEC GUIDE 7.5

Dedicated software guidance for NAWIs under NAWID; addresses software-related compliance requirements.

[Source ↗](#)

WELMEC GUIDE 7.2

Software guidance covering security, authenticity, integrity and validation principles for regulated measuring systems.

[Source ↗](#)

WHAT WE ADOPT

Configurable R-76 rules • explainable verdicts • maker-checker roles • audit trail • searchable records.

3 TECHNICAL DESIGN BASIS

OIML D 31:2023

General requirements for software-controlled measuring instruments: legally relevant software, security, data/parameter integrity and verification guidance.

AUDIT TRAIL

Interventions such as parameter changes should be recorded so they can be verified afterward.

DESIGN RESPONSE

SmartNAWI keeps rule IDs, observations, calculations and verdict history linked in one evidence chain.

[Source ↗](#)

OIML R 76 — UNDER REVISION

OIML lists revision of R 76 as an active high-priority project; a configurable rule layer makes future rule updates easier to absorb.

[Source ↗](#)

Live Demo | Working Prototype : <https://sachinthoke009.github.io/SIH-2026/>

Video Explanation :

For Detailed Project Please
check our prototype and video

